**P1**

(a) eq(7.45) reads:

$$\langle x|n\rangle = \frac{1}{\sqrt{n!}}\left(\sqrt{\frac{m\omega}{2\hbar}}\right)^n\left(x-\frac{\hbar}{m\omega}\frac{d}{dx}\right)^n\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}e^{-\frac{m\omega}{2\hbar}x^2}$$

$$\langle x|2\rangle = \frac{1}{\sqrt{2!}}\left(\frac{m\omega}{2\hbar}\right)\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\left(x-\frac{\hbar}{m\omega}\frac{d}{dx}\right)^2 e^{-\frac{m\omega}{2\hbar}x^2} =$$

$$= \underbrace{\frac{1}{2\sqrt{2}\,\pi^{1/4}}\left(\frac{m\omega}{\hbar}\right)^{1+\frac{1}{4}}}_{C}\left(x-\frac{\hbar}{m\omega}\frac{d}{dx}\right)\left(x-\frac{\hbar}{m\omega}\frac{d}{dx}\right)e^{-\frac{m\omega}{2\hbar}x^2} =$$

$$= C\left(x-\frac{\hbar}{m\omega}\frac{d}{dx}\right)\underbrace{\left[x\,e^{-\frac{m\omega}{2\hbar}x^2}-\frac{\hbar}{m\omega}\left(-\frac{m\omega}{2\hbar}\cdot 2x\right)e^{-\frac{m\omega}{2\hbar}x^2}\right]}_{2x\,e^{-\frac{m\omega}{2\hbar}x^2}}$$

$$= 2C\left\{x^2 e^{-\frac{m\omega}{2\hbar}x^2}-\frac{\hbar}{m\omega}\frac{d}{dx}\left(x\,e^{-\frac{m\omega}{2\hbar}x^2}\right)\right\} =$$

$$= 2C\left\{x^2 e^{-\frac{m\omega}{2\hbar}x^2}-\frac{\hbar}{m\omega}e^{-\frac{m\omega}{2\hbar}x^2}-\frac{\hbar x}{m\omega}\left(-\frac{m\omega}{2\hbar}\cdot 2x\right)e^{-\frac{m\omega}{2\hbar}x^2}\right\}$$

$$= \frac{\cancel{2}}{\cancel{2}\sqrt{2}\,\pi^{1/4}}\left(\frac{m\omega}{\hbar}\right)^{1+\frac{1}{4}}\left(2x^2-\frac{\hbar}{m\omega}\right)e^{-\frac{m\omega}{2\hbar}x^2} =$$

$$= \frac{1}{\sqrt{2}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\left(2\frac{m\omega}{\hbar}x^2-1\right)e^{-\frac{m\omega}{2\hbar}x^2}$$

which is exactly eq. (7.47) as it must be.

(b) $n=2 \Rightarrow \varepsilon = 5 \Rightarrow \frac{a_2}{a_0} = -2$ but $\frac{a_4}{a_2} = 0$

$$\Rightarrow h_{even}(y) = a_0 (1-2y^2)$$

$$\Rightarrow \psi_2(y) = a_0 (1-2y^2) e^{-y^2/2}$$

Since $y = \sqrt{\frac{m\omega}{\hbar}}\, x = \frac{x}{\ell}$ ; $\ell = \sqrt{\frac{\hbar}{m\omega}}$

$$\psi_2(x) = a_0 \left(1 - 2\frac{x^2}{\ell^2}\right) e^{-\frac{x^2}{2\ell^2}}$$

Find $a_0$: $1 = \int_{-\infty}^{\infty} dx\, |a_0|^2 \left(1 - 2\frac{x^2}{\ell^2}\right)^2 e^{-\frac{x^2}{\ell^2}} = \left(\frac{x}{\ell} = y\right) =$

$$= |a_0|^2 \cdot \ell \int_{-\infty}^{\infty} dy\, (1-2y^2)^2 e^{-y^2} =$$

$$= \ell\, |a_0|^2 \int_{-\infty}^{\infty} dy\, (1 - 4y^2 + 4y^4) e^{-y^2}$$

From $J_0(a) = \int_{-\infty}^{\infty} dx\, e^{-ax^2} = \sqrt{\frac{\pi}{a}}$

we find $J_1(a) = \int_{-\infty}^{\infty} dx\, x^2 e^{-ax^2} = -\frac{d}{da} J_0(a)$

$$J_2(a) = \int_{-\infty}^{\infty} dx\, x^4 e^{-ax^2} = +\frac{d^2}{da^2} J_0(a)$$

$$\Rightarrow J_1(a) = -\frac{d}{da}\sqrt{\frac{\pi}{a}} = -\sqrt{\pi}\left(-\frac{1}{2}\right) a^{-3/2} = \frac{1}{2}\sqrt{\frac{\pi}{a^3}}$$

$$J_2(a) = \frac{d^2}{da^2} J_0(a) = -\frac{\sqrt{\pi}}{2}\frac{d}{da} a^{-3/2} = \frac{\sqrt{\pi}}{2}\frac{3}{2} a^{-5/2} =$$

$$= \frac{3}{4}\sqrt{\frac{\pi}{a^5}}$$

Thus $\quad J_0(a=1) = \int_{-\infty}^{\infty} dy\, e^{-y^2} = \sqrt{\pi}$

$$J_1(a=1) = \frac{1}{2}\sqrt{\pi} \quad ; \quad J_2(a) = \int_{-\infty}^{\infty} dy\, y^4 e^{-y^2} = \frac{3}{4}\sqrt{\pi}$$

$\Rightarrow$ Normalization reads

$$1 = \ell\, |a_0|^2 \left(\sqrt{\pi} - 4\,\frac{\sqrt{\pi}}{2} + 4\,\frac{3\sqrt{\pi}}{4}\right) = 2\sqrt{\pi}\,\ell\, |a_0|^2$$

$$\Rightarrow |a_0| = \frac{1}{\sqrt{2\sqrt{\pi}\,\ell}} = \frac{1}{\sqrt{2}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}$$

$$\Rightarrow \psi_2(x) = \frac{1}{\sqrt{2}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \cdot$$

$$\cdot \left(1 - \frac{2m\omega}{\hbar}x^2\right) e^{-\frac{m\omega x^2}{2\hbar}}$$

which is exactly eq. (7.47) (up to a minus sign $= e^{i\pi}$)
just a global phase!

P2 $\psi_\alpha(x) = B \exp\left[-\frac{m\omega}{2\hbar}x^2 + \sqrt{\frac{2m\omega}{\hbar}}\,\alpha x\right]$

(a) Normalization

$$1 = |B|^2 \int_{-\infty}^{\infty} dx \; e^{-\frac{m\omega}{2\hbar}x^2 + \sqrt{\frac{2m\omega}{\hbar}}\alpha^* x} \; e^{-\frac{m\omega}{2\hbar}x^2 + \sqrt{\frac{2m\omega}{\hbar}}\alpha x}$$

Introduce dimensionless variable $y = \sqrt{\frac{m\omega}{\hbar}}\,x \equiv \frac{x}{l_0}$, $l_0 = \sqrt{\frac{\hbar}{m\omega}}$

$$\Rightarrow 1 = |B|^2 \sqrt{\frac{\hbar}{m\omega}} \int_{-\infty}^{\infty} dy \; e^{-y^2 + \sqrt{2}\alpha^* y + \sqrt{2}\alpha y} =$$

$$= |B|^2 l_0 \int_{-\infty}^{\infty} dy \; e^{-\left[\left(y - \frac{\alpha^*+\alpha}{\sqrt{2}}\right)^2 - \left(\frac{\alpha^*+\alpha}{\sqrt{2}}\right)^2\right]} = \left(y = z + \frac{\alpha+\alpha^*}{\sqrt{2}}\right)$$

$$= |B|^2 l_0 \, e^{\frac{(\alpha^*+\alpha)^2}{2}} \underbrace{\int_{-\infty}^{\infty} dz \; e^{-z^2}}_{\sqrt{\pi}}$$

$$\Rightarrow |B|^2 l_0 \sqrt{\pi}\, e^{\frac{(\alpha^*+\alpha)^2}{2}} = 1 \quad \Rightarrow B = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{(\alpha^*+\alpha)^2}{4}}$$

$$B = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-(\mathrm{Re}(\alpha))^2}, \quad \mathrm{Re}(\alpha) = |\alpha|\cos\varphi.$$

(b) To calculate $\langle x\rangle(t)$ we calculate $\langle x\rangle(0)$ and then obtain $t \neq 0$ result by $\alpha \to \alpha e^{-i\omega t}$ substitution.

$$\langle x\rangle = |B|^2 \int_{-\infty}^{\infty} dx \; x \, e^{-\frac{m\omega}{\hbar}x^2 + \sqrt{\frac{2m\omega}{\hbar}}(\alpha+\alpha^*)x} =$$

$$= \left(x = \sqrt{\frac{\hbar}{m\omega}}\, y = l_0 y\right) = |B|^2 l_0^2 \int_{-\infty}^{\infty} dy \; y \; e^{-y^2 + \sqrt{2}(\alpha+\alpha^*)y}$$

$$= |B|^2 l_0^2 \int_{-\infty}^{\infty} dy \; y \; e^{-\left(y - \frac{\alpha^*+\alpha}{\sqrt{2}}\right)^2 + \left(\frac{\alpha^*+\alpha}{\sqrt{2}}\right)^2} =$$

$$= \left(y = z + \frac{\alpha+\alpha^*}{\sqrt{2}}\right) = |B|^2 l_0^2 \, e^{(\alpha^*+\alpha)^2/2} \int_{-\infty}^{\infty} dz \left(z + \frac{\alpha+\alpha^*}{\sqrt{2}}\right) e^{-z^2}$$

$$= |B|^2 l_0^2 \, e^{\frac{1}{2}(\alpha+\alpha^*)^2} \, \frac{\alpha+\alpha^*}{\sqrt{2}} \cdot \sqrt{\pi}$$

Using result from (a) that $|B|^2 l_0 \sqrt{\pi}\, e^{\frac{(\alpha^*+\alpha)^2}{2}} = 1$

we obtain $\langle x \rangle = \frac{l_0}{\sqrt{2}}(\alpha+\alpha^*) = \sqrt{\frac{2\hbar}{m\omega}}\,|\alpha|\cos\varphi$

$$\Rightarrow \langle x\rangle(t) = \sqrt{\frac{2\hbar}{m\omega}}\,|\alpha|\cos(\varphi-\omega t)$$

because $\alpha = |\alpha| e^{i\varphi} \rightarrow \alpha e^{-i\omega t} = |\alpha| e^{i(\varphi-\omega t)}$

Next $\langle \hat{p}\rangle = (-i\hbar)\int dx\, |B|^2 e^{-\frac{m\omega}{2\hbar}x^2+\sqrt{\frac{2m\omega}{\hbar}}\alpha^* x}\cdot \frac{d}{dx} e^{-\frac{m\omega}{2\hbar}x^2+\sqrt{\frac{2m\omega}{\hbar}}\alpha x}$

$$= -i\hbar |B|^2 \int dx\, e^{-\frac{m\omega}{2\hbar}x^2+\sqrt{\frac{2m\omega}{\hbar}}\alpha^* x}\left(-\frac{m\omega}{\hbar}x+\sqrt{\frac{2m\omega}{\hbar}}\alpha\right)\cdot$$

$$\cdot e^{-\frac{m\omega}{2\hbar}x^2+\sqrt{\frac{2m\omega}{\hbar}}\alpha x} = (-i\hbar)|B|^2\int dx\left(-\frac{m\omega}{\hbar}x+\sqrt{\frac{2m\omega}{\hbar}}\alpha\right)\cdot$$

$$\cdot e^{-\frac{m\omega}{\hbar}x^2+\sqrt{\frac{2m\omega}{\hbar}}(\alpha+\alpha^*)x}$$

Now again $x = l_0 y = \sqrt{\frac{\hbar}{m\omega}}\, y$:

$$\langle p\rangle = i\hbar |B|^2 l_0 \int_{-\infty}^{\infty} dy \left(\frac{y}{l_0} - \frac{\sqrt{2}\alpha}{l_0}\right) e^{-y^2+\sqrt{2}(\alpha+\alpha^*)y} =$$

$$= i\hbar |B|^2 \int_{-\infty}^{\infty} dy\,(y-\sqrt{2}\alpha)\, e^{-\left(y-\frac{\alpha^*+\alpha}{\sqrt{2}}\right)^2+\left(\frac{\alpha+\alpha^*}{\sqrt{2}}\right)^2} =$$

$$= \left(y = z + \frac{\alpha+\alpha^*}{\sqrt{2}}\right) = i\hbar |B|^2 e^{\frac{(\alpha+\alpha^*)^2}{2}} \int_{-\infty}^{\infty} dz\left(z+\frac{\alpha+\alpha^*}{\sqrt{2}}-\sqrt{2}\alpha\right)\cdot$$

$$\cdot e^{-z^2}$$

$$= i\hbar |B|^2 e^{\frac{(\alpha+\alpha^*)^2}{2}} \int_{-\infty}^{\infty} dz\left(z+\frac{\alpha^*-\alpha}{\sqrt{2}}\right)e^{-z^2} =$$

$$= i\hbar |B|^2 e^{\frac{(\alpha+\alpha^*)^2}{2}} \frac{\alpha^*-\alpha}{\sqrt{2}}\sqrt{\pi} = i\hbar\frac{\alpha^*-\alpha}{\sqrt{2}\, l_0}$$

$$\alpha = |\alpha| e^{i\varphi} \Rightarrow \langle p\rangle(t=0) = i\hbar\frac{|\alpha|(-2i\sin\varphi)}{\sqrt{2}\sqrt{\frac{\hbar}{m\omega}}} =$$

$$= \sqrt{2m\omega\hbar}\,|\alpha|\sin\varphi$$

$$\Rightarrow \langle\hat{p}\rangle(t) = \sqrt{2m\omega\hbar}\,|\alpha|\sin[\varphi-\omega t].$$

P3

Bauer–Hausdorff formula:

for any operators $\hat{A}$ and $\hat{B}$ such that

$[\hat{A}, [\hat{A}, \hat{B}]] = [\hat{B}, [\hat{B}, \hat{A}]] = 0$ the following is true

$$e^{\hat{A}} e^{\hat{B}} = e^{\hat{A}+\hat{B}} e^{\frac{1}{2}[\hat{A}, \hat{B}]} \quad (*)$$

(Note that this implies $e^{\hat{A}} e^{\hat{B}} = e^{\hat{B}} e^{\hat{A}} e^{[\hat{A}, \hat{B}]}$.)

$$e^{\hat{A}+\hat{B}} = e^{\hat{A}} e^{\hat{B}} e^{-\frac{1}{2}[\hat{A}, \hat{B}]}$$

~~Now observe th~~

(a) $$\Rightarrow e^{\alpha \hat{a}^+ - \alpha^* \hat{a}} = e^{\alpha \hat{a}^+} e^{-\alpha^* \hat{a}} e^{-\frac{1}{2}[\alpha \hat{a}^+, -\alpha^* \hat{a}]} =$$

$$= e^{\alpha \hat{a}^+} e^{-\alpha^* \hat{a}} \underbrace{e^{\frac{1}{2}\alpha \cdot \alpha^* [\hat{a}^+, \hat{a}]}}_{e^{-\frac{1}{2}|\alpha|^2}}$$

Thus

(b) $$|\alpha\rangle = e^{\alpha \hat{a}^+ - \alpha^* \hat{a}} |0\rangle = e^{-\frac{1}{2}|\alpha|^2} e^{\alpha \hat{a}^+} e^{-\alpha^* \hat{a}} |0\rangle$$

but $\hat{a}|0\rangle = 0 \Rightarrow e^{-\alpha^* \hat{a}} |0\rangle = |0\rangle$

$$\Rightarrow |\alpha\rangle = e^{-\frac{1}{2}|\alpha|^2} \underbrace{e^{\alpha \hat{a}^+} |0\rangle}_{\sum_{k=0}^{\infty} \frac{\alpha^k}{k!} (\hat{a}^+)^k |0\rangle}$$

But $\frac{(\hat{a}^+)^k}{\sqrt{k!}} |0\rangle = |k\rangle$

(c) $$\Rightarrow |\alpha\rangle = e^{-\frac{1}{2}|\alpha|^2} \sum_{k=0}^{\infty} \frac{\alpha^k}{\sqrt{k!}} |k\rangle$$

just as we derived it before!

(d) $$\hat{a}\,|\alpha\rangle = \hat{a}\, e^{-|\alpha|^2/2} \sum_{m=0}^{\infty} \frac{\alpha^m}{\sqrt{m!}}\, |m\rangle =$$

$$= \left(\hat{a}\,|m\rangle = \sqrt{m}\,|m-1\rangle\right) = e^{-|\alpha|^2/2} \sum_{m=1}^{\infty} \frac{\alpha^m}{\sqrt{m!}} \sqrt{m}\,|m-1\rangle =$$

because $\hat{a}|0\rangle = 0$

$$= e^{-|\alpha|^2/2} \sum_{m=1}^{\infty} \frac{\alpha^m}{\sqrt{(m-1)!}}\, |m-1\rangle = (m = n+1) =$$

$$= e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^{n+1}}{\sqrt{n!}}\, |n\rangle = \alpha \cdot e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}}\, |n\rangle$$

$$= \alpha \cdot |\alpha\rangle$$

Thus $\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle$ indeed.